Block-Insure: Hybrid Real–Synthetic Evaluation of an Auditable Decentralized Healthcare Insurance Protocol

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***Abstract*—Healthcare claims combine confidential evidence, off-chain provider facts, probabilistic risk, human judgment, and funded settlement; no public dataset represents all five as ground truth. This paper presents Block-Insure, an Ethereum-compatible insurance protocol evaluated with 558,211 public-secondary Medicare claims plus eight controlled claim/registry attacks. Contracts enforce versioned policy economics, exact commit–reveal oracle consensus, deterministic auditor review, and pull-payment settlement, while AES-GCM evidence remains off-chain and Merkle roots bind external records. On 1,085 held-out providers, a calibrated Bernoulli model achieved ROC-AUC 0.8797 and PR-AUC 0.3576; logistic regression performed better at the selected operating point, limiting the claim to provider-risk prioritization. On 109,397 paired claims, combined rules and probabilistic routing detected all 54,705 injected attacks at 7.7881% untouched escalation. A 5,000-record runtime retained complete automatic and manual settlement workflows. Contract, reserve, evidence, gas, and load results expose both operational feasibility and bounded limitations. The contribution is a reproducible hybrid evaluation architecture, not proof of real-world claim-fraud detection.**

***Index Terms*—blockchain insurance, healthcare claims, hybrid dataset, oracle consensus, risk screening, confidential evidence**

I. Introduction

A healthcare claim is simultaneously an evidence, external-data, decision, and settlement problem. Blockchain can make state and money transitions auditable, but cannot independently prove that an off-chain medical event occurred—the reason authenticated and decentralized oracle mechanisms remain a separate trust layer [1], [2]. A model can prioritize attention, but cannot turn a weak provider label into confirmed claim fraud. Direct on-chain medical-document storage also creates confidentiality, cost, and permanence problems. Block-Insure therefore separates authority: contracts govern versions, roles, decisions, reservations, and balances; oracle workers report bound external observations; models advise routing; auditors resolve unsafe automation; and encrypted bytes remain off-chain.

Evaluation is equally fragmented. Prior healthcare-insurance systems have demonstrated blockchain adjudication and rule-based antifraud controls [3]–[6], while real Medicare studies show both the value and severe imbalance of provider-risk learning [7], [8]. Public claims provide realistic providers, beneficiaries, amounts, diagnoses, procedures, and time distributions, but not controlled truth for invoice reuse, altered amounts, registry mismatch, oracle conflict, stale versions, or proof corruption. Fully synthetic data provides those truths but weakens realism. This work combines both without conflating their labels.

The study asks whether (RQ1) real-derived aggregates rank weakly labeled providers under provider-held-out evaluation; (RQ2) explicit integrity rules and probabilistic screening complement each other on controlled attacks; (RQ3) a hybrid runtime preserves purchase-to-settlement functionality; and (RQ4) full-protocol evaluation reveals security, performance, evidence, and reserve boundaries hidden by model metrics alone.

The contributions are: (C1) a modular protocol for versioned policy economics, exact oracle consensus, confidential evidence, manual adjudication, appeals, and funded pull settlement; (C2) a provenance-preserving adapter for 558,211 public-secondary claims with distinct provider-risk, controlled-attack, and on-chain truth; (C3) provider-disjoint and paired-attack evaluations with calibration, comparators, ablations, and uncertainty; and (C4) contract, mutation, browser, Merkle/gas, evidence, reserve, and load experiments. Claims are bounded to the frozen local configuration; production decentralization, clinical truth, regulatory compliance, and universal fraud detection are outside scope.

II. Background and Related Work

Mackey *et al.* prototyped a blockchain antifraud framework for multi-stakeholder healthcare-claim adjudication, while identifying realistic empirical validation as future work [3]. Block-HI subsequently organized healthcare-insurance fraud into 12 scenarios and mapped them to blockchain controls, but did not learn from real claims or test an executable end-to-end settlement pipeline [4].

B-SAHIC implemented permissioned health-insurance processing with Hyperledger Fabric and measured transaction performance, but its predefined validation logic did not combine probabilistic risk, confidential evidence, and reserve-aware

settlement [5]. ChainCare demonstrated an Ethereum/React insurance DApp and gas-oriented prototype evaluation; external-data/oracle validation remained outside its realized trust model [9]. Islam *et al.* combined Fabric, IPFS, anonymous credentials, and smart-contract fraud rules, while leaving data-driven detection as an extension [6].

For analytics, Kaur *et al.* combined explainable machine learning with blockchain-backed healthcare-fraud records, reporting strong ranking performance on large claims data [8]; Hancock *et al.* showed that imbalance handling and explainability materially affect Medicare provider-risk modelling [7]. Neither evaluates adversarial registry/oracle truth together with a complete claim lifecycle. FraudAuditor instead supports expert investigation of collusive health-insurance relationships, reinforcing the value of human review rather than forced autonomous classification [10].

Thus, existing work separately covers blockchain adjudication, fraud rules, provider-risk learning, and expert investigation. Block-Insure addresses the composition gap: it preserves distinct weak-label, controlled-attack, and protocol truth; binds oracle observations to versions; and evaluates model, evidence, contract, reserve, and browser behavior in one reproducible workflow.

III. System and Data Architecture

A. Authority and Lifecycle

The React client authenticates by wallet signature and performs evidence cryptography. Express orchestrates authenticated requests, projections, indexing, analytics, and notifications; MongoDB is not economic authority. Two workers independently reconstruct registry snapshots and submit version-bound observations, treating external data as an explicit oracle boundary [1], [2]. Seven Solidity modules divide protocol state: `InsuranceManager` controls claim transitions; `PolicyEconomics` snapshots rules and reserves coverage; `OracleCoordinator` enforces commit–reveal; `ClaimAdjudicator` owns assignment, votes, decisions, and the payout vault; `EvidenceRegistry` anchors identities and tree heads; benefits and deployment registries isolate auxiliary state.

A purchase snapshots terms and opens coverage. Submission validates ownership, incident, amount, documents, duplicate keys, and available coverage before reserving liability. An exact positive oracle quorum allocates payout; false, conflict, or timeout becomes eligible for review. Four auditors are assigned deterministically: three valid votes approve, two invalid votes reject, and expiry rejects. Insufficient treasury creates a visible `FUNDING_REQUIRED` liability; after funding, claimant withdrawal produces `SETTLED`. Appeals increment claim version so old oracle or decision records cannot finalize the new cycle.

B. Evidence, Registry, and Failure Containment

The browser encrypts with AES-GCM using claim/version/uploader/type as authenticated data [11], uploads ciphertext, and wraps the 256-bit key for proxy transformation; related PRE/blockchain work likewise separates protected health data from the ledger [12]. The backend need not learn plaintext or the key. A per-claim chain binds document order; an append-only RFC6962-style evidence tree supports inclusion and consistency proofs [13]; the chain anchors authenticated heads, not medical bytes, consistent with hybrid off-chain/on-chain storage principles [14]. Separately, oracle requests snapshot the healthcare-registry version/root, worker membership, quorum, and model identity. Missing model artifacts, disagreement, timeout, insufficient funds, unavailable storage, and stale projections fail without granting silent backend settlement authority.

TABLE I
Hybrid Partitions and Controlled Injection

Quantity	Train	Calibration	Test
Claims	343,537	105,277	109,397
Providers	3,420	905	1,085
Injected attacks	40,969	12,600	54,705
Injection rate	≈12%	≈12%	50%

C. Hybrid Dataset and Runtime

The implemented source is the compatible four-file Kaggle *Healthcare Provider Fraud Detection Analysis* Medicare benchmark [15]: provider labels, beneficiaries, inpatient claims, and outpatient claims joined only by native `Provider` and `BeneID`. It contains 558,211 claims (40,474 inpatient; 517,737 outpatient), 5,410 providers, and 138,556 beneficiaries. The public `PotentialFraud` field is a weak provider label, not claim truth; this interpretation follows provider-oriented Medicare fraud research rather than relabeling every claim [7].

For provider p and seed 20260828, the first 32 bits of $H(\texttt{split:}\|s\|p)$ assign 64%/16%/20% hash buckets to train/calibration/test. All claims of a provider remain together (Table I). Temporal features use only prior events; treatment amount ranges and 95th-percentile velocity thresholds are learned from training. Beneficiaries can cross partitions and are disclosed as a limitation.

Independent claim-ID hashes select eight attacks: used/cancelled invoice, inflated amount, date mismatch, blacklisted provider, near duplicate, provider compromise, and noisy record. Untouched and attacked test rows remain paired by source claim. Synthetic augmentation is established for imbalanced healthcare-fraud evaluation [16]; here it is used more narrowly for controlled causal attacks. Thus *provider risk*, *controlled attack truth*, and *on-chain protocol truth* remain separate targets inside one compatible schema.

Offline studies use all applicable claims. Runtime uses an order-independent bottom-hash sample of 4,995 real-derived training records plus five stable normal/attack presets, producing 5,000 unique invoices and a depth-13 Merkle tree. The original claim amount is reimbursement plus deductible; demonstration value is $\max(amount/10{,}000, 0.0001)$, a local scale rather than currency conversion. A manifest records hashes, counts, seed, and version `blockinsure-hybrid-v1.2.0`.

IV. Methodology

A. Provider-Risk Estimation

For provider p with claims C_p, the real-data model aggregates

$$\mathbf{g}_p = \left[n_p, u_p, \bar{A}_p, A_p^{\max}, \bar{I}_p, \bar{P}_p, \bar{R}_p, \bar{H}_p, \bar{V}_p\right], \tag{1}$$

representing claim/unique-beneficiary volume, mean/maximum amount, inpatient/procedure/repeat ratios, mean chronic count, and activity-anomaly ratio. Training-only quartiles binarize each dimension:

$$x_{pj} = \mathbb{1}\left[g_{pj} \geq Q_{0.75,j}^{train}\right]. \tag{2}$$

Bernoulli Naive Bayes (BNB) with Laplace $\alpha = 1$ computes

$$\ell_y(\mathbf{x}) = \log \pi_y + \sum_j [x_j \log \theta_{jy} + (1 - x_j) \log(1 - \theta_{jy})],$$

$$\theta_{jy} = \frac{N_{jy} + \alpha}{N_y + 2\alpha}, \qquad P_y = \frac{e^{\ell_y}}{\sum_{k\in\{0,1\}} e^{\ell_k}}. \tag{3}$$

Calibration providers fit Platt parameters (a, b) using a held-out sigmoid mapping [17],

$$\hat{p} = \sigma\left(a \log \frac{P_1}{1 - P_1} + b\right), \tag{4}$$

and select threshold $t^* = \arg\max_{t\in\{.01,\ldots,.99\}} F_1(t) = 0.35$. Comparators are majority negative, unweighted feature count, and unweighted/class-weighted logistic regression. Thresholds never use test labels. Three leave-group-out ablations and 1,000 seeded provider bootstraps estimate contribution and uncertainty [18].

B. Authority-Aware Claim Routing

The separate frozen protocol model consumes 18 registry/claim features. For missed-attack, false-escalation, and review costs (C_m, C_f, C_r), asymmetric cost-sensitive triage [19] defines the probability crossover and expected evaluation cost

$$t_e = \frac{C_r + C_f}{C_m + C_f}, \qquad t_m = \max(t_e, t_{capacity}), \tag{5}$$

$$J = C_m FN + C_f FP_{legit} + C_r N_{review}. \tag{6}$$

With $(10, 1, 0.25)$ and nominal review capacity 20%, $t_e = 0.113636$, $t_{capacity} = 0.217969$, and $t_m = 0.217969$. Let $H(\mathbf{x})$ denote hard comparison failure and $A(\mathbf{x})$ committed/fuzzy anomaly. Routing is

$$\mathcal{R}(\mathbf{x}, \hat{p}) = \begin{cases} \textsc{Reject-Oracle}, & H(\mathbf{x}) = 1, \\ \textsc{Manual-Review}, & A(\mathbf{x}) = 1 \ \vee \ \hat{p} \geq t_m, \\ \textsc{Auto-Verify}, & \text{otherwise.} \end{cases} \tag{7}$$

Probability can abstain to review but cannot create authoritative rejection. This bounded abstention follows the broader principle that a model may defer uncertain decisions to a human agent [20]. Paired model-only, rule-only, and combined evaluations isolate each attack's causal effect.

An oracle request binds query q, claim version c, registry version/root (r, ρ), and model m inside worker result h. Commit/reveal binds a salt s, while exact consensus counts

$$\kappa_o = H(request, c, r, v, h, m, s_o),$$
$$d = H(v, h, c, r, m). \tag{8}$$

Two equal digests finalize; conflict or timeout cannot pay and routes conservatively.

C. Settlement, Reserve, and Evidence Experiments

For claim amount A, deductible rate/cap (d, D), and insurer share s (basis points), integer settlement is

$$q = \min\left(A, \left\lfloor Ad/10^4 \right\rfloor, D\right),$$
$$L = \left\lfloor (A - q)s/10^4 \right\rfloor, \quad R = A - L, \tag{9}$$

subject to $reserved + settled + L \leq coverage$. Protected treasury is $T_{min} = U + B + \lfloor Er/10^4 \rfloor$ for unfunded liability U, buffer B, exposure E, and reserve ratio r.

Reserve simulation separates claim frequency and severity, following standard non-life modelling structure [21]: $K \sim Poisson(Nf)$ and $X_i = \exp(\ln m + \sigma Z_i)$. Setting $A_i = \min(X_i, 1)$, let L_i be the contract liability obtained from (9); annual loss is $S = \sum_{i=1}^{K} L_i$. For capital $C = Er$ on exposure E,

$$P_{short} = \frac{1}{M} \sum_{j=1}^{M} \mathbb{1}[S_j > C], \quad ES = E[S - C \mid S > C]. \tag{10}$$

Evidence tests measure AES-GCM, proxy transformation, and Merkle proof latency/size plus eight tamper/availability/access failures. Contract evaluation combines unit, stateful invariant, mutation, coverage, browser workflow, gas, bytecode, registry, and local concurrency experiments.

D. Experimental Protocol and Metrics

Experiments ran on a local x64 Windows host (Node.js 24.16.0). Solidity 0.8.26 targeted Cancun with IR and one optimizer run on Hardhat 2.28.0, chain 31337. Two workers used exact quorum two and 25-block commit/reveal windows; manual review used four seats with 3-valid/2-invalid thresholds. The provider model trained/calibrated/tested on 3,420/905/1,085 providers. Logistic comparators used 1,200 iterations, learning

Methodology Diagram 1: Hybrid Evaluation Pipeline
Public claim/beneficiary/provider files → provider-disjoint split → aggregate risk model → paired deterministic attacks → offline and runtime evaluation

Methodology Diagram 2: Authority-Aware Decision Pipeline
Hard integrity comparison and calibrated probability → auto-verify or human review → exact oracle consensus → reserve, fund, and pull settlement

Fig. 1. Reserved spaces for the two principal methodology diagrams: hybrid-data construction/evaluation and authority-aware claim routing/settlement.

rate 0.08, and L2 penalty 0.002. Reserve evaluation used 5,000 periods, seed 410034, 250 bootstrap resamples, and a 1% shortfall target; evidence used 30 repetitions.

Provider/routing outcomes report precision $TP/(TP+FP)$, recall $TP/(TP+FN)$, $F_1 = 2PR/(P+R)$, MCC, ROC-AUC, average-precision PR-AUC, Brier $N^{-1}\sum_i (p_i - y_i)^2$, and ten-bin calibration error. PR-AUC is emphasized because positive-class evaluation is especially informative under imbalance [22]. Performance measures include p50/p95 latency, gas, deployed bytes, proof size, and successful local submissions; no local rate is interpreted as public-chain throughput. A reserved deployment protocol will reuse frozen bytecode on Hardhat, Sepolia, and Polygon Amoy and archive receipt gas, native-token fee, confirmation time, address, and transaction hash. Reserved Slither analysis will pin analyzer/compiler versions, export machine-readable findings, triage by severity and detector, remediate confirmed defects, and rerun; neither pending experiment is treated as a result. Semantic reproducibility identities bind data/model/code/compiler artifacts; wall-clock timestamps are excluded.

V. Results

A. Provider Risk

On 1,085 held-out providers, BNB produced TP=41, TN=914, FP=87, FN=43 (Table II). Its ROC-AUC was 0.8797 (95% provider-bootstrap CI 0.8488–0.9078), PR-AUC 0.3576 (0.2631–0.4599), and F1 0.3868 (0.3011–0.4706). Majority accuracy was 0.9226 but recall/F1 were zero, illustrating imbalance. Both logistic comparators improved recall, F1, MCC, and ranking; BNB is therefore a frozen, interpretable primary artifact rather than the best observed classifier. Removing volume/activity caused the largest ranking loss (ΔROC=-0.1015, ΔPR=-0.1463). Removing coarse setting/clinical fields improved F1 by 0.0755, indicating redundancy/noise in this encoding rather than clinical irrelevance.

TABLE II
Held-Out Provider Models and Feature-Group Ablations

Model	Prec.	Recall	F1	MCC	ROC	PR
Frozen BNB	.3203	.4881	.3868	.3324	.8797	.3576
Majority negative	0	0	0	0	.5000	.0774
Rule score	.3047	.4643	.3679	.3110	.8727	.3175
Logistic	.4000	.5476	.4623	.4156	.8978	.3857
Weighted logistic	.3932	.5476	.4577	.4107	.8987	.3966
BNB w/o vol./activity	.2581	.5714	.3556	.3075	.7781	.2114
BNB w/o financial	.4531	.3452	.3919	.3520	.8682	.3525
BNB w/o setting/clinical	.4000	.5476	.4623	.4156	.8945	.3706

B. Paired Attack Routing

On 109,397 paired test claims, combined routing produced TP=54,705, TN=50,466, FP=4,226, FN=0, ROC-AUC 0.9571, PR-AUC 0.9207, and mean injected probability increase 0.2168. Table III shows why: deterministic rules dominate explicit mismatches but detect only 8.09% of provider compromise; the model covers provider compromise but misses inflated and noisy records. Combination reaches every deterministic scenario with only 0.0036 percentage-point more untouched escalation than rules. The 100% figure is scenario-bounded, not real-fraud accuracy.

TABLE III
Routing Ablation on Paired Claims

Strategy	Untouched	Recall	Prec.	F1	MCC
Model only	.0037%	55.25%	99.99%	71.17%	.6177
Rules only	7.7845%	88.74%	92.00%	90.34%	.8107
Combined	7.7881%	100%	92.83%	96.28%	.9255

TABLE IV
Controlled-Attack Recall by Scenario

Scenario	n	Model	Rules	Combined
Blacklisted provider	6,940	10.46%	100%	100%
Cancelled record	6,999	100%	100%	100%
Date mismatch	6,789	31.51%	100%	100%
Inflated amount	6,879	0%	100%	100%
Near duplicate	6,874	100%	100%	100%
Noisy record	6,737	0%	100%	100%
Provider compromise	6,702	100%	8.09%	100%
Used invoice	6,785	100%	100%	100%

C. Whole-Protocol Measurements

Table V converts cross-layer outcomes into a single numerical matrix. All verification groups passed and all eight valid sampled mutants were killed, but 65.07% branch coverage remains a material gap. Five browser role/smoke tests and two purchase-to-`SETTLED` workflows passed: one exact-oracle path and one negative-oracle/manual-vote path. Root anchoring saved 99.62% storage gas at 500 records, although it does not pay off-chain availability cost. At concurrency 50, 11 nonce-queue failures exposed local saturation. The manager's 314-byte headroom under Ethereum's 24,576-byte code-size limit [23] motivates further modularization.

D. Pending Deployment and Static Security

Table VI reserves the final reporting structure for two experiments that are not yet executed. Dashes are intentionally blank and must be replaced only by archived deployment receipts and a reproducible Slither run; they are not zero values. Testnet-token fees will be reported in native units, with any dated USD equivalent labeled as an estimate. Slither findings will be separated from confirmed defects and will not be presented as a formal audit.

TABLE VI
Reserved Deployment and Slither Measurements

(a) Multi-network contract deployment

Network	Suite	Gas	Fee	Conf.
Hardhat local	7/7	–	Free	–
Sepolia (L1)	–	–	–	–
Amoy (L2)	–	–	–	–

(b) Slither finding triage

Level	Found	Valid	Fixed	Left
High	–	–	–	–
Medium	–	–	–	–
Low	–	–	–	–
Informational	–	–	–	–

E. Reserve Adequacy

At the configured 5% ratio, only baseline met the 1% target (Table VII). High-frequency and combined stress failed every simulated year; heavy-tail failure was 28.38%. Grid recommendations increased to 8–15%, showing that a fixed software default is not robust capital policy.

TABLE V
WHOLE-SYSTEM NUMERICAL RESULTS MATRIX

Experiment	Primary measurements	Interpretation boundary
Verification suites	Contract 127; backend 63; oracle 13; hybrid adapter 8; all passed	Tests, not formal proof
Solidity assurance	Statements 93.77%; branches 65.07%; functions 90.59%; lines 94.39%; mutants killed 8/8	Sampled mutants; branch gap remains
Browser workflows	Smoke/role 5/5 in 30.3 s; complete workflows 2/2 in ≈84 s	Frozen local stack
5,000-leaf registry	Depth 13; cold build 90.089 ms; proof build/verify mean .0040/.0246 ms; valid 100/100	Cached local snapshot
Merkle storage gas	500 hashes 11,402,190 gas; root 43,851; saving 99.62%	Storage only
Contract size	Manager 24,262 bytes; EIP-170 headroom 314 bytes	Maintainability risk
Claim load	$c = 20$: 20/20, 48.38 claims/s, p95 412.74 ms; $c = 50$: 39/50, 49.29/s, p95 790.66 ms	Hardhat/one machine
AES-GCM	p95 encrypt/decrypt ms: 16 KiB .057/.056; 256 KiB .477/.622; 1 MiB 3.432/3.341	Excludes network
Proxy re-encryption	Capsule 932 bytes; transform/decrypt p95 15.48/19.48 ms (worst tested row)	Per recipient
Evidence proof/security	512 leaves: 288 bytes, build/prove p95 8.743 ms, verify p95 .025 ms; attacks detected 8/8	Bounded cases

TABLE VII
RESERVE RESULTS AT CURRENT 5% CAPITAL

Scenario	P_{short} [95% CI]	ES (ETH)	Ratio*
Baseline	.22% [.10,.34]	.6076	5%
High frequency	100% [100,100]	25.7306	10%
Heavy tail	28.38% [27.34,29.43]	3.3047	8%
Combined stress	100% [100,100]	44.0556	15%

*Smallest tested ratio meeting 1% target.

VI. DISCUSSION AND LIMITATIONS

A. Interpretation

RQ1 is supported only as prioritization: held-out ROC-AUC remains materially above chance, but moderate threshold precision/recall and stronger logistic comparators reject any claim of BNB superiority or autonomous fraud judgment. RQ2 is supported within the controlled benchmark: rules and probability fail on different attacks, and the authority-aware union closes those gaps. RQ3 is supported by both automatic and auditor-routed settlements on the 5,000-record hybrid runtime. RQ4 is supported by negative as well as positive evidence: stress reserve failure, nonce saturation, manager bytecode pressure, and lower branch coverage expose constraints that model AUC alone cannot reveal.

Table VIII compresses the literature gap without implying universal superiority. Block-Insure's distinction is the evaluated composition of real-derived provider risk, controlled attack truth, version-bound oracle consensus, confidential evidence, and funded human fallback. It remains partial in explainability and has no completed public multi-network deployment.

TABLE VIII
FEATURE COMPARISON WITH CLOSEST HEALTHCARE SYSTEMS

Capability	[4]	[6]	[8]	[9]	Ours
Real claims/provider data	–	–	Y	–	Y
Controlled attack truth	P	P	–	–	Y
Learned risk model	–	–	Y	–	Y
Version-bound multi-oracle	–	–	–	–	Y
Confidential evidence	–	Y	–	–	Y
Human fallback/appeal	–	P	–	–	Y
Funded reserve settlement	–	–	–	P	Y
Model explainability	–	–	Y	–	P
Public multi-network run	–	–	–	P	TBD

Y: demonstrated; P: partial/conceptual; –: not reported; TBD: pending.

The central design decision is epistemic separation inside a unified schema. Real-derived fields supply realistic distributions and weak provider risk; deterministic mutations supply measurable claim-level attack truth; contracts supply protocol truth. Evaluation occurs at the valid unit for each target. This avoids both a fabricated cross-source join and the misleading phrase "real fraud claims."

Authority follows evidence strength. A cryptographic/registry comparison can reject oracle verification; probability can only abstain to review; exact oracle agreement can trigger settlement; conflict/timeout cannot; auditors act through bounded assignment and quorum; and the claimant withdraws only a funded, calculated allocation. The result is defense in depth, not replacement of human or institutional trust.

B. Validity and Security Boundaries

Construct validity: `PotentialFraud` is inherited weak provider risk, not legal or claim-level fraud. The eight deterministic attacks exercise known fields and may be easier than adaptive behavior; 100% combined recall is therefore scenario coverage. Local settlement proves workflow execution, not clinical truth.

Internal validity: provider-disjoint splitting, training-only thresholds, calibration-only operating points, paired claims, and frozen seeds reduce leakage. However, beneficiaries cross partitions, Boolean features lose information, and comparators omit boosted, graph, and sequential models. Two oracle processes use separate accounts/snapshots but share one project operator; quorum collusion and admin-controlled registry publication remain threats.

External validity: the public-secondary Medicare-style source may not transfer across countries, insurers, coding systems, or time. Only 5,000 records enter runtime; all 558,211 are used offline where applicable. Hardhat automining, local MongoDB/IPFS substitutes, one Windows host, and absent CPU/RAM capture prevent production throughput claims. Evidence timings exclude browser/network storage. Poisson-lognormal reserve scenarios omit dependence, inflation, expenses, premiums, reinsurance, and regulation.

Assurance/privacy: manifests bind source/output hashes, seeds, models, compiler, bytecode, and semantic results, but independent reproduction requires the same source files. Tests, 65.07% branch coverage, and eight mutants are not formal verification or a professional audit; formal model checking has been applied to blockchain claim-processing workflows and remains complementary future assurance [24]. Browser key storage is not production custody; revocation cannot retract recovered plaintext; access metadata and availability remain

observable risks. No public-testnet deployment, independent hospital oracle, penetration/privacy assessment, HIPAA/GDPR certification, or statutory reserve claim is made.

VII. Conclusion

Block-Insure demonstrates a functioning insurance protocol whose evaluation does not force one dataset to impersonate every truth source. On held-out providers, the frozen BNB reached ROC-AUC 0.8797 and PR-AUC 0.3576; combined routing detected all 54,705 injected test attacks with 7.7881% untouched escalation. Encrypted evidence and Merkle snapshots bind external material, while contracts enforce versioned consensus, review, reservations, funding, and withdrawal. Two end-to-end paths settled successfully, yet 65.07% branch coverage, 314-byte manager headroom, concurrency-50 failures, and reserve stress shortfalls expose meaningful limits.

Future work should add independent artifact replication; beneficiary-disjoint and stronger provider models; stochastic, coordinated, poisoning, replay, and oracle attacks; independently governed workers/registries; public-testnet cost and latency; robust relaying; formal invariant analysis and professional audit; production key custody/storage; and qualified actuarial calibration. Until then, the contribution is a bounded, reproducible architecture for auditable irregularity screening and settlement, not production insurance or universal fraud detection.

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